



Technical Service Bulletin

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QST30 Industrial - Fuel Leak - Turbocharger Oil Supply Hose Rubbing Number 4 Left Bank Injector Fuel Supply Tube

QST30 Industrial - Fuel Leak - Turbocharger Oil Supply Hose Rubbing Number 4 Left Bank Injector Fuel Supply Tube

Core Issue

An injector fuel supply tube leak has been reported on QST30 industrial engines. The fuel leak is caused by rubbing or chaffing contact between the left bank turbocharger oil supply hose, Part Number 3087969, and the number 4 injector fuel supply tube, Part Number 3093373. Reported leaks have occurred in less than 1000 hours of engine operation.

Confirmation

Inspect the lines when performing service on an industrial QST30 engine to make sure of adequate clearance between the fuel supply tubes and the braided turbocharger oil supply lines.

Wear, if noted between the left bank turbocharger oil supply hose, Part Number 3087969, and the number 4 injector fuel supply tube, Part Number 3093373, is unacceptable.

Resolution

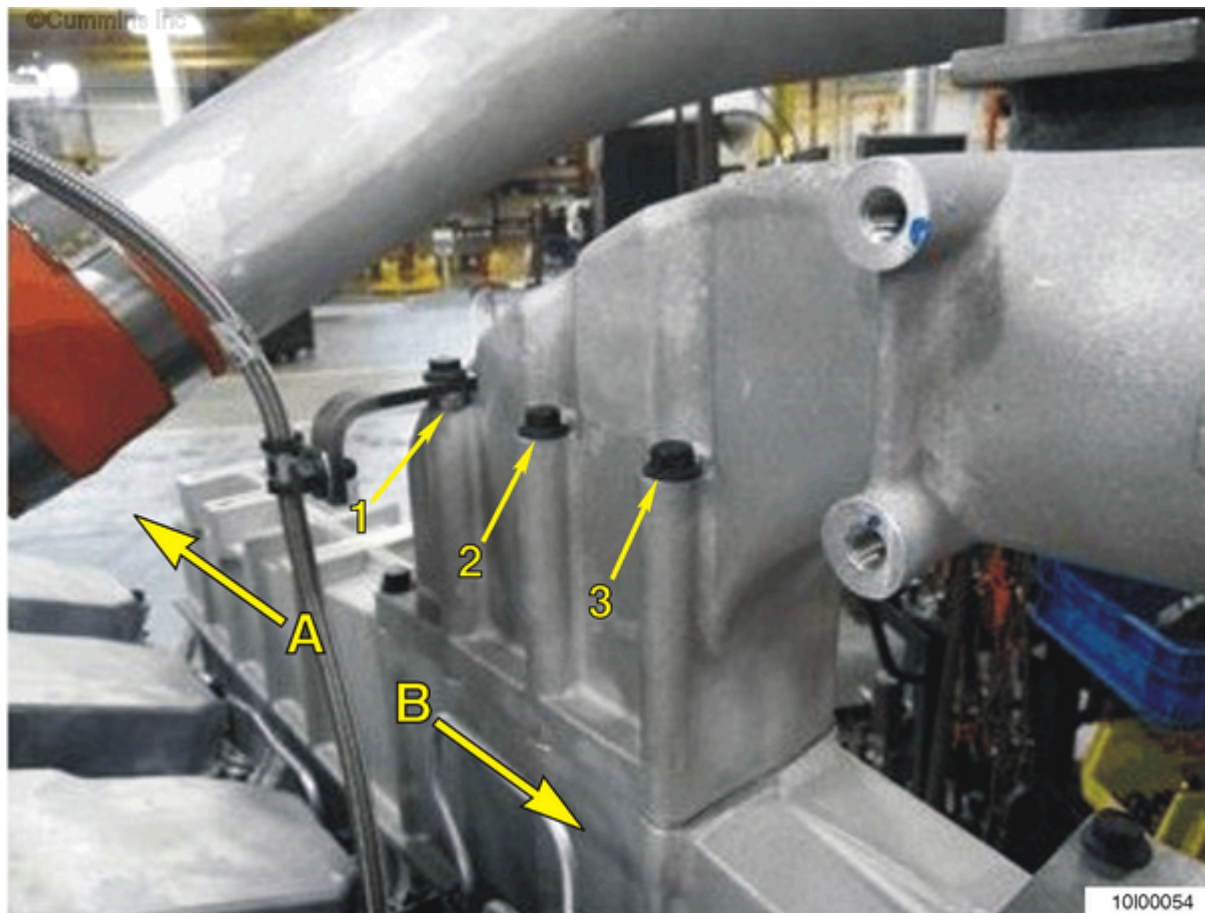
Make sure there is ample clearance between the turbocharger oil supply hose and the injector fuel supply tube. A P-clip may need to be rotated 180 degrees from the current orientation to make sure of proper clearance between the turbocharger oil supply hose and the injector fuel supply tube. If wear is observed between the turbocharger oil supply hose and the injector fuel supply tube, both should be replaced.

Make sure that either Part Number 3094824, 100 mm bracket, or Part Number 4371605, 60 mm bracket, is installed on the engine with the following turbocharger plumbing options:

TP Option	Air Crossover Bolt Hole Location	Turbocharger
TP5705	RB Middle	Front Out
TP5709	LB Rear, RB Rear	Front Out
TP5710	LB Rear, RB Rear	Front Out
TP5711	LB Middle	Rear Out

TP Option	Air Crossover Bolt Hole Location	Turbocharger
TP5715	LB Middle	Rear Out
TP5716	RB Middle	Front Out
TP5717	LB Middle	Rear Out
TP5718	LB Rear, RB Rear	Front Out
TP5719	LB Middle	Rear Out
TP5721	LB Middle	Rear Out
TP5722	RB Middle	Front Out
TP5759	LB Rear, RB Rear	Front Out
LB: Left Bank		
RB: Right Bank		

Engines with these TP options, built on or after 22-May-2014, have the 60 mm bracket installed.
ESN first is 37261832.



Air Crossover Bolt Hole Location.

1. Front capscrew hole.
 2. Middle capscrew hole.
 3. Rear capscrew hole.
- A. Front of engine.
B. Rear of engine.

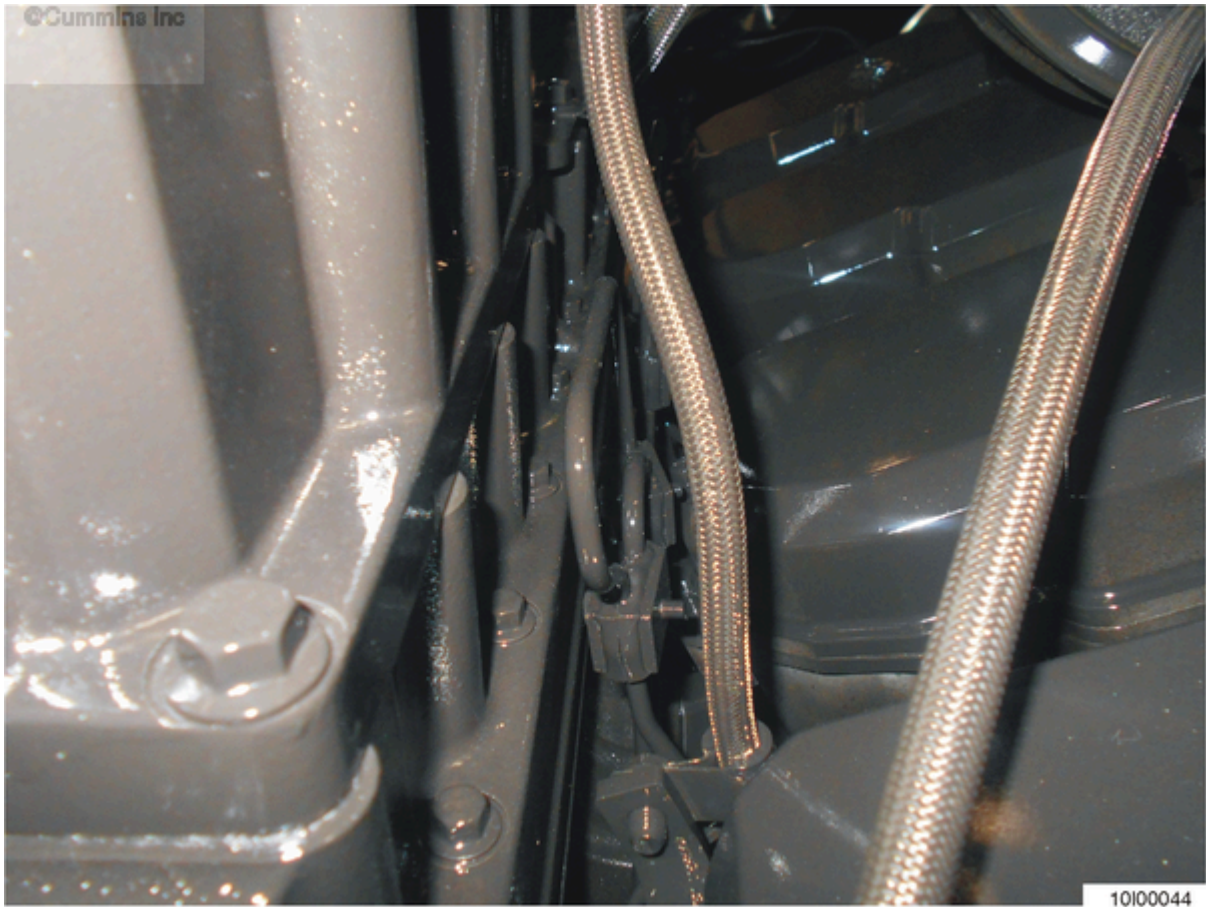


Reported wear.

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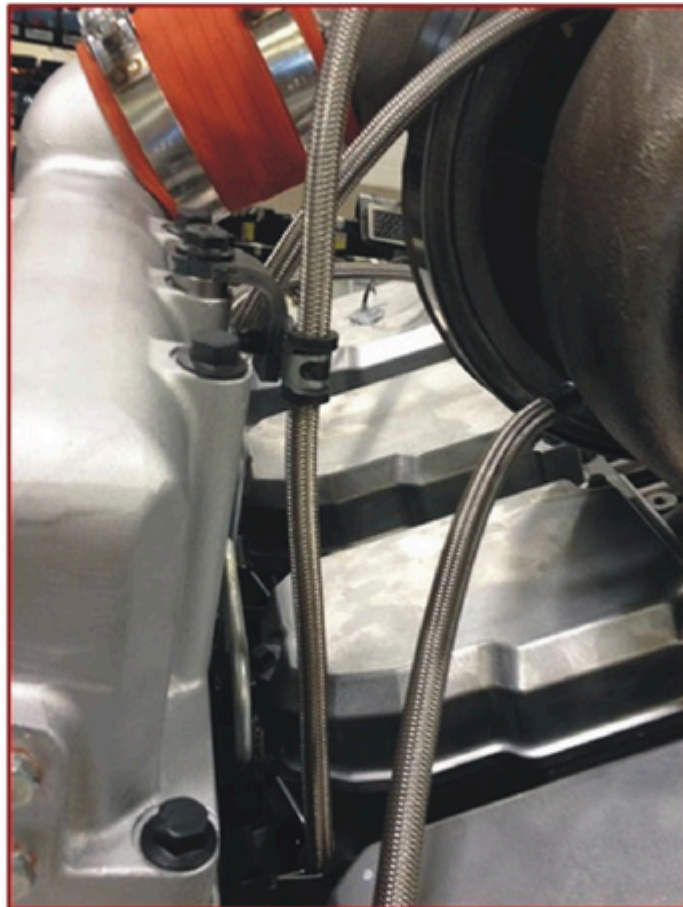


Looking down on the left bank of the engine.



Proper line routing making sure the turbocharger oil supply hose does **not** rub against the injector fuel supply tube or the drain tube. (Viewed forward from the rear of the engine.)

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Rotate the P-clip 180 degrees toward the rear of the engine to improve clearance between the turbocharger oil supply hose and fuel supply tube.

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10J00212

Looking down on the left bank of the engine, after rotating P-clip.

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06I00011

60 mm bracket, Part Number 4371605, installed on engine, ESN first 37261832.

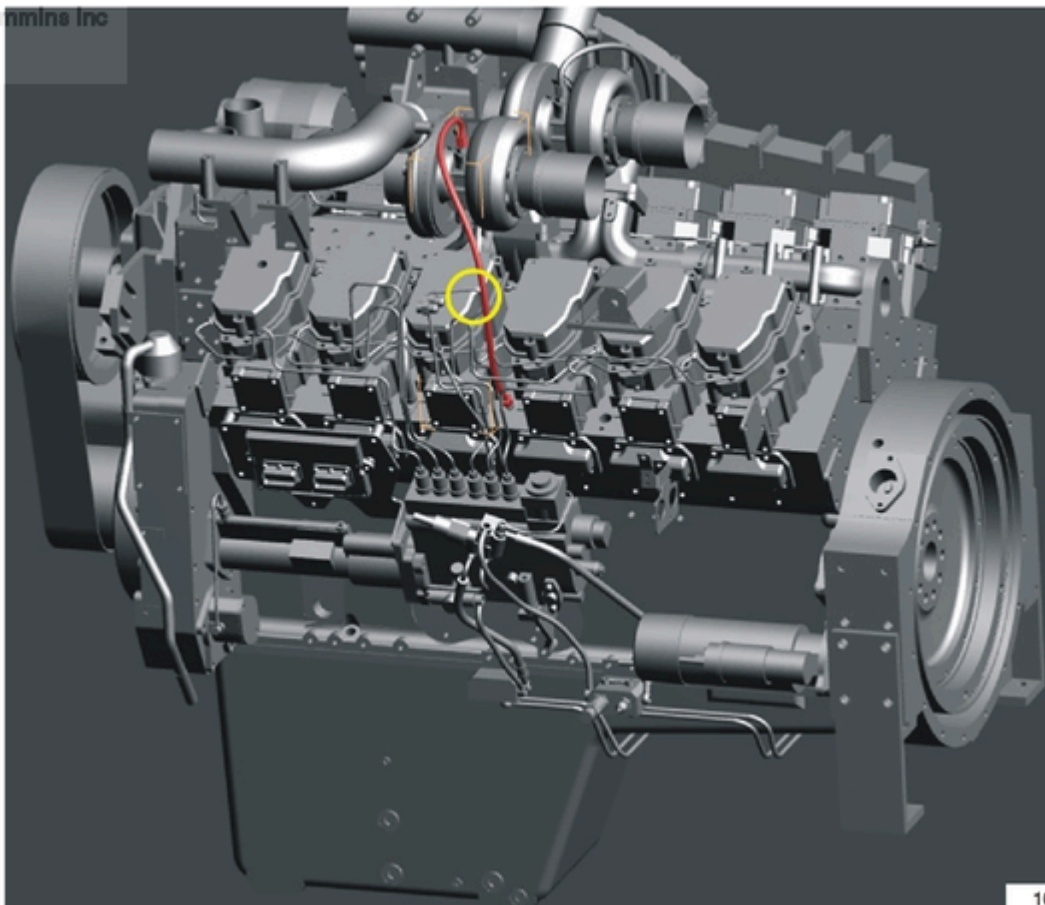
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06I00012

100 mm bracket, Part Number 3094824, installed on engine.

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Location of the turbocharger oil supply hose rubbing on the fuel supply tube.

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Document History

Date	Details
2014-5-5	Module Created
2014-7-1	none
2014-8-24	none

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